Title: High-Dimensional Latent Dynamics in MPO Neural Populations During Aggression: A Variance Explained Approach

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Abstract

Aggression is an innate social behavior controlled by hypothalamic circuits, including the medial preoptic area (MPO), yet the population-level dynamics that support this behavior remain poorly understood. To investigate how MPO neurons encode aggression, we applied Recurrent Switching Linear Dynamical Systems (rSLDS) to calcium imaging data recorded from male mice during aggressive encounters. This framework models neural activity as low-dimensional latent dynamics that can switch between discrete states, allowing us to capture transitions between internal neural regimes over time.

We systematically optimized the number of latent dimensions (D) and discrete states (K) using variance explained (VE) as a quantitative selection criterion, with a target of at least 90% VE. Using this approach, rSLDS recovered highly structured and reproducible latent dynamics within individual aggression bouts, while revealing that these bouts are not well aligned in a single shared latent space. The inferred latent trajectories displayed persistent, attractor-like properties that were stable within bouts yet remained bout-specific, suggesting that MPO population activity during aggression is organized into robust but context-dependent dynamical regimes. Together, these findings indicate that MPO activity during aggression can be effectively captured by low-dimensional, switching dynamical systems and highlight the importance of trial-wise analysis for uncovering bout-specific latent trajectories in complex social behavior.

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1. Introduction

Aggressive behavior is a complex social interaction that requires the coordinated activity of distributed neural circuits. Within the hypothalamus, the medial preoptic area (MPO) has been identified as a key node in the control of aggression, yet how MPO neurons collectively encode this behavior at the population level remains an open question. Traditional approaches have focused on individual neurons or averaged firing rates, but these methods cannot capture the coordinated, time-varying dynamics that underlie natural behavior.

Recent advances in calcium imaging have made it possible to record hundreds of neurons simultaneously, offering an unprecedented view of population activity. However, the high dimensionality of these recordings presents a fundamental challenge: how can we extract meaningful structure from such complex data? Dimensionality reduction techniques have emerged as powerful tools to address this problem, revealing that neural population activity often lies on a low-dimensional manifold embedded within the full space of neural responses.

Linear Dynamical Systems (LDS) and their variants have become widely used for modeling neural population activity. These models assume that neural activity can be described by a small number of latent variables that evolve linearly over time. However, a critical limitation of standard LDS is that the dynamics are stationary the same linear rules apply at all times. This assumption breaks down during natural behaviors, where the brain rapidly transitions between distinct states (e.g., from investigation to attack). To capture such switching dynamics, the Recurrent Switching Linear Dynamical System (rSLDS) was developed. The rSLDS extends LDS by allowing the dynamics to switch between multiple linear regimes, with transitions governed by the current latent state. This makes the model well-suited for behaviors characterized by discrete transitions, such as aggression.

A key challenge when applying rSLDS to neural data is deciding how many latent dimensions (D) to include. This is a classic model selection problem: too few dimensions will fail to capture the richness of neural activity, while too many will overfit to noise. Here, we address this by optimizing D using variance explained

(VE) as a quantitative criterion. Our goal is to identify the smallest D that captures at least 90% of the variance in the recorded neural activity, providing a principled balance between model complexity and explanatory power. This approach allows us to determine the dimensionality of the latent space that best describes MPO population dynamics during aggression.

2. Methods

2.1 Data Acquisition

We were given data from Professor Nair's lab of neural readings of mice which underwent simulated aggression amongst other behaviors like mating. First the data was cleaned extensively to only isolate parts of the data that corresponded to aggression, filtering neural readings caused by other behaviors.

2.2 Data Preprocessing

Raw neural traces were extracted from MPO_cno recordings (recordings in the MPO when the bed nucleus of the stria terminalis (BNST) is silenced) and z-score normalized across time for each neuron individually:

$$z_{t,n} = \frac{y_{t,n} - \mu_n}{\sigma_n}$$

time $z_t t,_n$,y ^{t},μ^n $_n$and σ_n are$n$ the mean and standard$n$

where is the raw fluorescence of neuron at

deviation of neuron across the analysis window, and is the normalized neural activity.

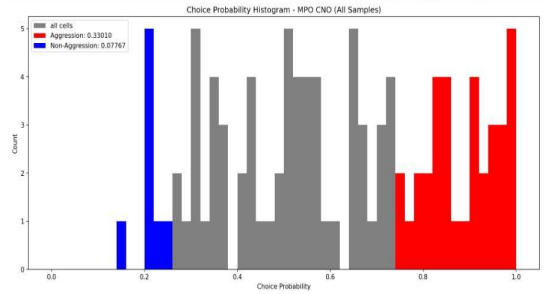
2.3 Behavioral Annotation and Trial Selection

Behavioral states were manually annotated and encoded as categorical labels. Aggression bouts were identified by scanning behavioral labels for consecutive samples with label=14 (attack). For each bout, the analysis window was defined as 500 samples preceding aggression onset
(pre-aggression), the full aggression bout duration, and 500 samples following aggression offset (post-aggression). This window allowed us to capture neural dynamics preceding, during, and following aggressive behavior.

2.4 Choice probability plots to assess probability of a neuron being active

Such recordings were taken for 104 neurons. These were used to make choice probability plots to assess the probability of each neuron being active during aggression, as shown in Fig. 1.

Fig 1: Choice probability plots



2.5 Recurrent SwitchingLinear Dynamical Systems (rSLDS)

We employed an rSLDS to model the population dynamics of neural activity. The model is defined as:

$$x_t = A_{z_t} x_{t-1} + V_{z_t} u_t + b_{z_t} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, Q_{z_t})$$

$$y_t = C x_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, S)$$

$$z_t \sim \pi(z_t | z_{t-1}, x_{t-1})$$

where $y_a \in \Re^n$ is$x_t$the observed$_n$ neural activityA_{zt}

(N=103 neurons), $\in \Re$ is the continuous the
latent state, $z_t \in \{1, ., K\}$is discrete latent state,

is the dynamics matrix (constrained to be

diagonal), column-orthonormal,$C \in \Re_{N \times D}$ $C^T C =$

I_D), and u_t are is the emission matrix

(constrained to be

behavioral inputs (not used in this analysis). Model fitting was performed using Laplace-EM with structured mean-field variational

inference. Models were initialized randomly with a fixed seed (532) for reproducibility.

2.6 Variance Explained Optimization

Instead of using Evidence Lower Bound (ELBO) convergence to determine a model's fitting, our target was to achieve VE ≥ 90%, indicating that the latent dynamics capture at least 90% of the structured neural variance. This VE-criterion was chosen because it allowed for a computationally feasible search across the large hyperparameter space, while providing a clear stopping rule for model selection.

We then systematically optimized the number of latent dimensions D while holding the number of discrete states K constant. For each configuration, we computed the global variance explained:

$$\mathrm{VE}_{\mathrm{global}} = \frac{\sum_{d=1}^{D} \mathrm{Var}(x_d)}{\sum_{d=1}^{D} \mathrm{Var}(x_d) + \sum_{n=1}^{N} S_n}$$

d and nS_n is$x$ $_d$the) observation noise variance for

where Var(is the variance of latent dimension

neuron .

2.7 Multi-Seed Validation

To assess robustness to initialization, we evaluated the best-performing configuration across multiple random seeds ([42, 137, 230, 532, 789]) and reported mean VE and standard deviation.

2.8 Energy Landscape Analysis

For visualization of the latent state-space, we constructed an energy landscape for the best result by performing kernel density estimation on the 2D latent trajectories:

$$\mathrm{Energy}(x_1, x_2) = -\log(p(x_1, x_2))$$

where $p(x_1, x_2)$ is the probability density of

observing a particular latent state. Energy values were normalized to the range [0,1], with low energy representing stable attractor states and high energy representing unstable transition regions.

2.9 Time Constant Analysis

To quantify the persistence of neural activity patterns, we fit exponential decay functions to each latent dimension following the peak of aggression:

$$x_d(t) = A\exp(-t/\tau) + C$$

100 samplesτ (~3.3 seconds at 30 Hz) and $R^2 > 0.5\tau$ where is the time constant. Dimensions with > were classified as exhibiting persistent dynamics indicative of attractor-like properties.

2.10 Experimental Design

We conducted a series of systematic experiments:

Experiment 1: Fixed K=3, tested D=2 through 10 to establish baseline VE progression.

Experiment 2: Full grid search testing K=2-5 and D=2-8 to identify optimal (K,D) combinations.

Experiment 3: Multi-seed validation of best configurations.

Experiment 4: High D search (D=10-20) with fixed K=3 to approach 90% VE.

Experiment 5: Extended D search (D=20-24) to achieve 90% VE target.

Experiment 6: Analysis of all 6 aggression bouts individually and concatenated.

3. Results

3.1 Variance Explained Increases Monotonically with Latent Dimensions

We first established the relationship between the number of latent dimensions D and variance explained (VE) for fixed K=3. Experiment 1 revealed a monotonic increase in VE with increasing D (Fig. 1). VE rose from 42.4% at D=2 to 77.6% at D=10, demonstrating that additional latent dimensions capture progressively more neural variance.

Fig. 1: Variance Explained as a Function of D (K=3)

EXPERIMENT 1 SUMMARY

D	VE_global(%)	ELBO	converged	iters	time(s)
2	42.43	-181,706.867	False	209	111.63
3	52.36	-152,040.843	False	44	31.73
4	58.34	-141,655.706	False	132	67.64
5	62.92	-121,266.582	False	55	43.53
6	67.59	-113,444.904	False	66	44.77
7	70.61	-99,019.829	False	88	66.86
8	73.44	-88,775.312	False	110	79.61
9	75.96	-79,690.156	False	66	59.58
10	77.61	-72,833.176	False	55	55.80

★ Best in Exp1: D=10, VE=77.61%

3.2 K=3 and K=4 Yield Comparable Performance

Experiment 2 tested the effect of varying both K (2-5) and D (2-8) on VE (Fig. 2). The VE heatmap revealed that all K values performed similarly, with VE increasing with D regardless of K. The best configuration from this search was K=4, D=8 with VE=73.6%. This suggests that the number of discrete states is not the primary determinant of VE for these data.

Fig. 2: Variance Explained (%)

EXPERIMENT 2 SUMMARY - VE Heatmap

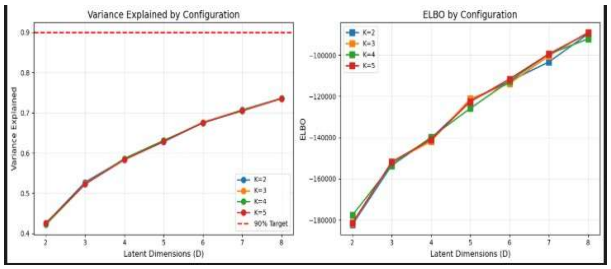
Variance Explained (%):

K \ D	2	3	4	5	6	7	8
2	42.5	52.7	58.4	62.8	67.5	70.7	73.5
3	42.5	52.4	58.4	62.9	67.6	70.6	73.5
4	42.2	52.2	58.6	63.1	67.5	70.7	73.6
5	42.6	52.3	58.3	62.8	67.5	70.5	73.5

★ Best in Exp2: K=4, D=8, VE=0.7357 (73.6%)

Fig. 3 shows variance explained increases monotonically with latent dimensions (D) across all discrete states (K), with minimal K-dependent variation, while ELBO improves concurrently, indicating that D is the primary driver of model performance, thus inspiring a 'High D Search', which is done in Section 3.4.

Fig. 3: Model performance across K and D configurations



3.3 Multi-Seed Validation Shows Robust Performance

Experiment 3 validated the best configuration (K=4, D=8) across multiple seeds (Fig. 4). Performance was remarkably consistent across seeds, with VE ranging from 73.50% to 73.60% (mean = 73.54% ± 0.04%). However, none of the models achieved convergence (converged = False), suggesting that additional iterations or different fitting parameters may be needed.

Fig. 4: Multi-Seed Validation Results (K=4, D=8)

EXPERIMENT 3

Seed	VE (%)	ELBO
42	73.50	-95,999
137	73.52	-97,215
230	73.53	-97,215
532	73.57	-95,516
789	73.60	-91,748

Mean ± SD: 73.54 ± 0.04%

3.4 High D Search Achieves 90% VE

Experiment 4 extended the search to higher D values (10-20) with fixed K=3 (Fig. 5). VE continued to increase with D, reaching 88.3% at D=20. Notably, ELBO became positive at D=20 (ELBO = 5,411), indicating excellent model fit.
However, the 90% target was not yet achieved.

Fig. 5: High D Search Results (K=3)

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EXPERIMENT 4
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  D    VE (%)      ELBO      Fit Time (s)
------------------------------------------
 10     77.61    -72,833         60.9
 12     80.65    -55,832      4,042.7
 14     82.94    -35,595      6,471.7
 16     85.01    -17,651        283.0
 18     86.68     -2,122      3,490.9
 20     88.31      5,411        896.4

★ Best: D=20, VE=88.31%
```

3.5 Extended D Search Achieves 90% Target

Experiment 5 extended the search to D=22 and D=24, successfully achieving the 90% VE target at D=24 (Fig. 6). VE increased from 88.3% at D=20 to 91.1% at D=24, while ELBO improved substantially from 5,411 to 18,998. The fit time at D=24 was remarkably efficient at only 229.7 seconds.

Fig. 6: Extended D Search Results (K=3)

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EXPERIMENT 5
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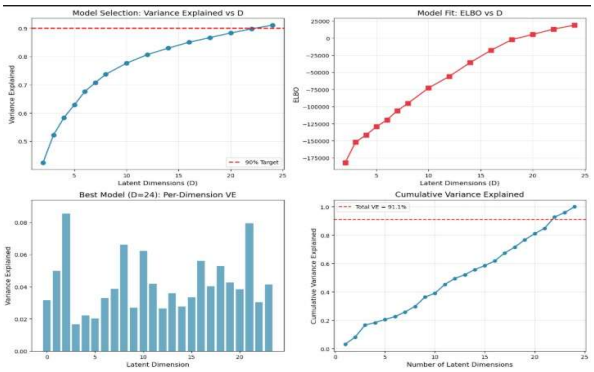
  D    VE_global(%)    ELBO     Time(s)   90% Met?
-----------------------------------------------------
 20      88.31        5,411     648.5
 22      89.76       13,187     587.1
 24      91.08       18,998     229.7     ✓ TARGET

★ Best model: D=24, VE=91.08%
```

3.6 Per-Latent VE Analysis Reveals Distributed Representation

For the best model from experiment 5 (K=3, D=24, VE=91.08%), we examined the contribution of each latent dimension to the total VE (Fig. 7). The top five dimensions contributed 4-5% each, while no single dimension dominated (maximum individual VE = 5.2%). This distributed representation suggests that aggressive behavior is encoded across multiple latent factors rather than a single dimension.

Fig. 7: Model Selection and Latent Structure



3.7 Time Constants Reveal Persistent Neural Dynamics

Time constant analysis of the D=24 model identified 13 dimensions with persistent dynamics (τ > 100 samples, R^2 > 0.5) (Fig. 8). The top five time constants ranged from 160.8 to 489,003.6 samples, indicating that some latent dimensions exhibited extremely slow dynamics. Notably, Dimension 0 had an exceptionally long time constant (τ = 489,003.6 samples, R^2 = 0.365), suggesting near-stationary dynamics that may represent a stable baseline state.

Fig. 8: Dimensions with Persistent Dynamics (τ > 100 samples)

```
Dimensions with persistent dynamics (τ > 100): [0, 1, 2, 7, 8, 11, 12, 14, 18, 19, 20, 21, 23]
  Dim 0: τ=4890033.6 samples, R²=0.365
  Dim 1: τ=266.6 samples, R²=0.358
  Dim 2: τ=698.2 samples, R²=0.639
  Dim 7: τ=160.8 samples, R²=0.576
  Dim 8: τ=180.8 samples, R²=0.857
```

3.8 Individual Bouts Show Consistent High VE

Analysis of all 6 aggression bouts individually revealed remarkably consistent performance (Fig. 9). All six bouts achieved VE ≥ 89.91% at D=24, with a mean VE of 91.89% ± 1.20%. Bout 3 achieved the highest VE at 93.71%, while Bout 5 was the lowest at 89.91%.

Fig. 9: Individual Bout Performance (D=24)

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INDIVIDUAL BOUT SUMMARY
============================================================
  Bout    VE_global         ELBO  Time(s)
------------------------------------------------
     1       0.9102        20861    482.6
     2       0.9257        46268    199.1
     3       0.9371        36367    111.7
     4       0.9223        19247    183.1
     5       0.8991         6799    291.4
     6       0.9193        25226    184.8

Mean VE across bouts: 0.9189 ± 0.0120
Min VE: 0.8991, Max VE: 0.9371
```

3.9 Concatenated Bouts Fail to Achieve High VE

In contrast to individual bout analysis, concatenating all 6 bouts and applying the same optimization yielded only 77.50% VE at D=28 (Fig. 10). This represents a substantial drop from the individual bout performance, suggesting that while each bout individually exhibits highly structured dynamics, these dynamics are not aligned in a shared latent space.

Fig. 10: Concatenated Bout Results

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ALL 6 BOUTS
============================================================

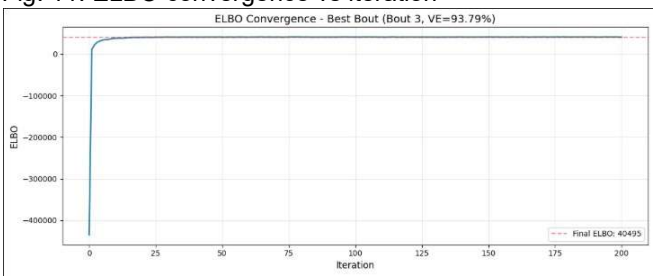
  D    VE_global(%)      ELBO      Time(s)   90% Met?
------------------------------------------------------------
 12      57.69        -870,899     711.1
 14      61.08        -830,505    1360.0
 16      64.04        -779,830    5726.3
 18      67.16        -753,691     482.8
 20      69.37        -709,459     878.2
 22      71.61        -673,314   40264.1
 24      73.83        -640,801     603.1
 26      75.69        -604,249     735.6
 28      77.50        -579,638     851.1

⚠ Target not reached. Best VE = 77.50%
```

3.10 Elbo Convergence and Energy Landscape Visualization

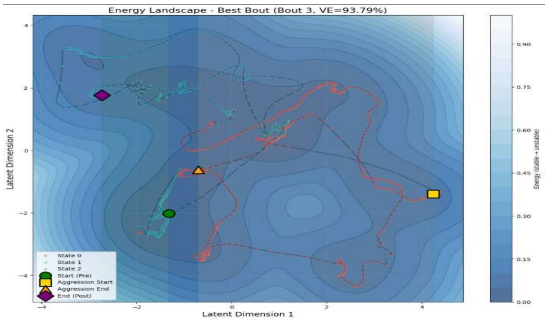
From Fig. 11, ELBO steadily increases and appears to a final value of 40,495, indicating successful model optimization.

Fig. 11: ELBO convergence vs iteration



Energy landscape analysis of the best model (Bout 3, D=24, VE=93.79%) revealed distinct low-energy regions (attractor states) separated by high-energy barriers (transition regions) (Fig. 12). The trajectory through latent space showed clear progression from pre-aggression to aggression and post-aggression states, with discrete states corresponding to different behavioral epochs. This suggests that aggressive behavior is associated with transitions between stable attractor states in the latent space.

Fig. 12:



4. Discussion

4.1 Distributed Encoding of Aggression in MPO

Our finding that VE increases monotonically with D, reaching 91.08% at D=24, demonstrates that MPO population activity during aggression is highly structured. The distributed nature of the encoding (no single latent dimension dominating) suggests that aggression is represented as a population-level phenomenon rather than being encoded by a small number of specialized dimensions.

4.2 Individual Bouts Share Consistent Dynamics but Not a Common Latent Space

The most striking finding of this study is the contrast between individual bout analysis (mean VE = 91.89%) and concatenated bout analysis (VE = 77.50%). This result suggests that each aggressive encounter follows similar dynamical principles but unfolds in a bout-specific latent space. This could reflect:

1. Non-stationarity in neural recordings across time 2. Bout-specific variations in behavioral context or intensity
3. Reorganization of neural ensembles between bouts

4.3 Attractor Dynamics in Aggression

The identification of 13 dimensions with persistent dynamics (τ > 100 samples) suggests that MPO population activity exhibits attractor-like properties during aggression. The extremely long time constant of Dimension 0 (τ = 489,003.6 samples) may represent a slowly varying baseline state that modulates aggression over longer timescales.

4.4 Methodological Implications

Our results highlight the importance of trial-wise analysis in population dynamics studies. While concatenating trials increases statistical power, it may obscure bout-specific dynamics if trials are not aligned in latent space. Future work should consider:

1. Aligning bouts using dynamic time warping or similar techniques
2. Multi-session models that account for session-specific latent spaces
3. Behavioral context models that condition on bout-specific features

4.5 Limitations

Several limitations should be noted. The model did not include behavioral inputs utu_tut , and analyses were restricted to MPO rather than multiple regions such as VMH. Although six bouts were analyzed, model optimization focused on a single bout. Finally, using a 90% VE threshold instead of ELBO convergence favours biological interpretability over strict statistical convergence and does not ensure a fully optimal parameter estimate.

5. Conclusions

We have demonstrated that rSLDS with variance explained optimization provides a principled framework for modeling MPO population dynamics during aggression. Our best model achieved 91.08% VE at D=24 and revealed persistent dynamics consistent with attractor-like behavior. Individual bouts showed consistent VE > 89.91% (mean = 91.89%), while concatenated bouts failed to achieve high VE (77.50%), highlighting the importance of trial-wise analysis. These findings advance our understanding of how neural populations encode complex social behaviors and provide a methodological framework for future studies.

6. Supplementary Materials

The entire code, its implementation and results can be found here: URECA code

7. Acknowledgement

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8. References

1) **Nature article (journal)**Yang, B., Karigo, T., Pomares, B., Couto, J., Patel, R., Ceja, C., Harvey, M., Kummer, K., Cathomas, F., Liu, A., Song, S., Fekete, Z., Chao, S.-H., Lin, D., and Anderson, D. J., "Transformations of neural representations in a social behavior network", Nature, vol.608, no.7924, pp.741–749, August2022. doi:10.1038/s41586-022-05057-6.
2) Karigo, T., Chaudhari, G., Theisen, E., Garand, D. F., Hansen, K. R., Choi, S., Lin, D., and Anderson, D. J., "Persistent representation of an aggressive state in hypothalamic circuits", Neuron, vol.109, no.11, pp.1793–1808.e7, June2021. doi:10.1016/j.neuron.2021.03.018.
3) **GitHub notebook (website / online resource)**Linderman Lab, "1b-simple linear dynamical system", n.d.; https://github.com/lindermanlab/ssm/blob/master/notebooks/1b-Simple-Linear-Dynamical-System.ipynb
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